

Qatar-10b

R-0084

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-10_231026 · created 2026-07-15T23:16:32+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2460243.6735 +/- 0.0024 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1736 +/- 0.0079
Transit depth (Rp/Rs)^2	3.01 +/- 0.27 [%]
Orbital Inclination (inc)	83.57 +/- 1.92
Airmass coefficient 1 (a1)	1.09 +/- 0.019
Airmass coefficient 2 (a2)	-0.0492 +/- 0.0085
Scatter in the residuals of the lightcurve fit is	1.76 %
Best Comparison Star	#1 - [181, 391]
Optimal Aperture	3.17
Optimal Annulus	8.46
Transit Duration (day)	0.111 +/- 0.01

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.1736 $\pm$ 0.0079 vs 0.1265 (deviation 5.96 σ)
Duration fit vs published	0.111 d vs 0.1154 d (deviation 0.44 σ)
Mid-time O-C	-21.5 min
Depth SNR	22.0
Residual scatter	1.76 %

Light curve

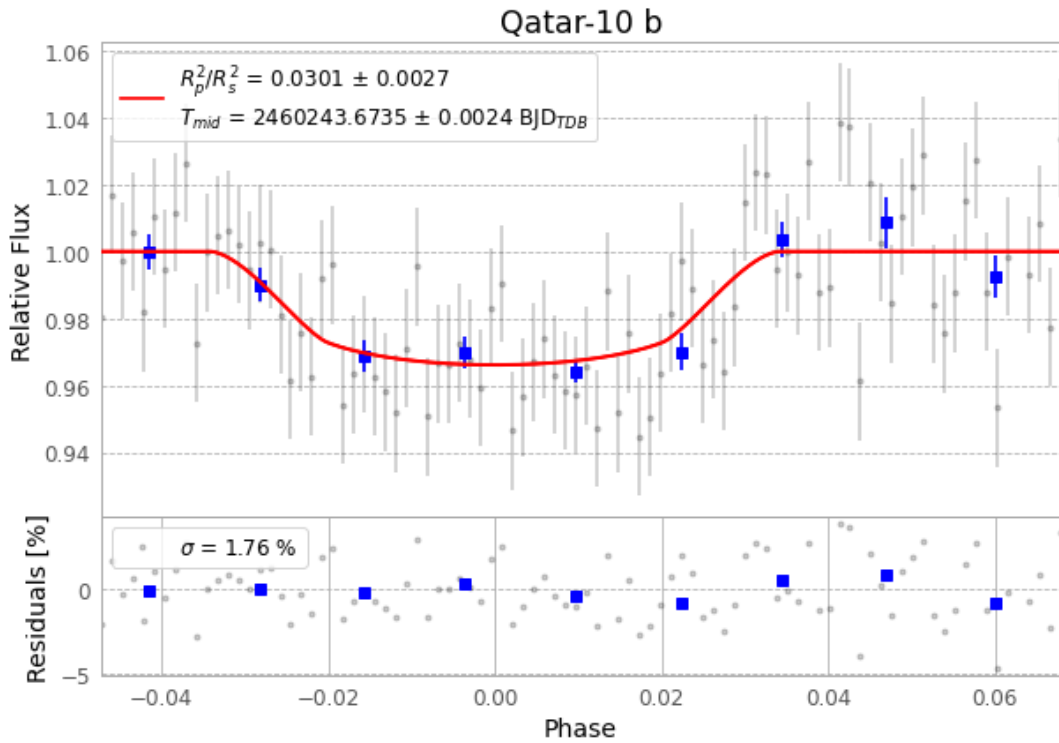


Figure 1 — FinalLightCurve_Qatar-10 b_2023-10-25.png (SHA-256 e6d029be4baece25...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [289, 192], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 1ccea9599d7d6160...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13

Data provenance

92 FITS frames (60 MB), Qatar-10231026021515.FITS → Qatar-10231026064816.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-10-231026.log (b1fd27245239...), FinalLightCurve_Qatar-10 b_2023-10-25.png (e6d029be4bae...), FinalLightCurve_Qatar-10 b_2023-10-25.csv (e67e4623daac...)

Compute resources

Wall time 135.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-15 23:16Z.